Inspiraton is initiated by the contraction of the diaphragm muscles and external intercostal muscles, which pulls the ribs and sternum upwards and outwards and increases the volume of the thoracic chamber in the dorso–ventral axis, forcing the lungs to expand the pulmonary volume. The increase in pulmonary volume and decrease in the intrapulmonary pressure forces the fresh air from outside to enter the air passages into the lungs to equalize the pressure. This process is called **inspiration.**

Relaxation of the diaphragm allows the diaphragm and sternum to return to its dome shape and the internal intercostal muscles contract, pulling the ribs downward reducing the thoracic volume and pulmonary volume. This results in an increase in the intrapulmonary pressure slightly above the atmospheric pressure causing the expulsion of air from the lungs. This process is called **expiration.**

On an average, a healthy human breathes 12–16 times/minute. An instrument called **Spirometer** is used to measure the volume of air involved in breathing movements for clinical assessment of a person's pulmonary function.



You are at high level in a mountain above the sea level. Suddenly you get palpitation and nausea. What condition are you suffering from? What are the other symptoms for this disease and how can it be reduced?

6.3.1 Respiratory volumes and capacities

The volume of air present in various phases of respiration is denoted as

Respiratory volumes: (Figure 6.5)

- **Tidal Volume (TV)** Tidal volume is the amount of air inspired or expired with each normal breath. It is approximately 500 mL., i.e. a normal human adult can inspire or expire approximately 6000 to 8000mL of air per minute. During vigorous exercise, the tidal volume is about 4–10 times higher.
- **Inspiratory Reserve volume (IRV)** Additional volume of air a person can inspire by forceful inspiration is called Inspiratory Reserve Volume. The normal value is 2500–3000 mL.
- **Expiratory Reserve volume (ERV)** Additional volume of air a person can forcefully exhale by forceful expiration is called Expiratory Reserve Volume. The normal value is 1000–1100 mL.
- **Residual Volume (RV)** The volume of air remaining in the lungs after a forceful expiration. It is approximately 1100–1200 mL.

Respiratory capacities:

- **Vital capacity (VC)** the maximum volume of air that can be moved out during a single breath following a maximal inspiration. A person first inspires maximally then expires maximally. VC = ERV+TV+IRV
- **Inspiratory capacity (IC)** The total volume of air a person can inhale after normal expiration. It includes tidal volume and inspiratory reserve volume. IC = TV+IRV
- **Expiratory capacity (EC)** The total volume of air a person can exhale after normal inspiration. It includes

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